

ASSIGNMENT-8

Formation of Matrices and vector using MATLAB and carrying out operations involving matrices :

Compute the following Questions

1. $2^5/(2^5-1)$ and compare with $(1-1/2^5)^{-1}$
2. $(3(\sqrt{5}-1)/(\sqrt{5}+1)^2)-1$
3. Area = πr^2 with $r = \pi^{1/3} - 1$
4. e^3 , $\ln(e^3)$, $\log_{10}(e^3)$ and $\log_{10}(10^5)$
5. $e^{\pi\sqrt{163}}$
6. Solve $3^x = 17$ for x and check the result. (The solution is $x=\ln 17/\ln 3$. You can verify the result by direct substitution.
7. $\sin\pi/6$, $\cos\pi$, and $\tan\pi/2$
8. $\sin^2\pi/6 + \cos^2\pi/6$
9. $y=\cosh^2x-\sinh^2x$ with $x=32\pi$
10. $(1+3i)/(1-3i)$
11. $e^{i\pi/4}$
12. $\exp(\pi/2*i)$ and $\exp(\pi/2i)$
13. The equation of a straight line is $y= m*x + c$, where m and c are constants. Compute the y co-ordinates of a line with slope $m=0.5$ and the intercept $c = -2$ at the following x -coordinates:
 $x = 0, 1.5, 3, 4, 5, 7, 9, \text{ and } 10$
14. Create a vector t with 10 elements: 1,2,...,10. Now compute the following quantities
 $x = t\sin(t)$
 $y = (t-1)/(t+1)$
 $z = \sin(t^2)/t^2$
15. All points with coordinates $x=r \cos\theta$ and $y=r \sin\theta$, where r is a constant, lie on a circle with radius r , i.e, they satisfy the equation $x^2+y^2=r^2$. Create a column vector for θ with the values $0, \pi/4, \pi/2, 3\pi/4, \pi, 5\pi/4$. Take $r=2$ and compute the column vectors x and y . Now check that x and y indeed satisfy the equation of a circle, by computing the radius $r = \sqrt{(x^2+y^2)}$